A counterexample to the explicit multiplicity–gap criterion

Conjecture 00000001421

Abstract

We construct a sequence of exactly-half-lazy, uniform Cayley random walks on finite abelian groups of increasing order. Its ordinary spectral gap is $1/N$. The smallest nontrivial transition eigenvalue has multiplicity $N-1$; the smallest positive eigenvalue of $I-P$ has multiplicity one. Consequently both possible multiplicity-to-gap ratios diverge. Nevertheless the actual total-variation distance at every positive integer time is $\frac{1}{2}(1-1/N)^t$. The true quarter-mixing time diverges, while its ratio to the quarter-to-three-quarter transition width tends to one. The family has no standard total-variation cutoff. This refutes the conjecture's explicit numerical sufficient condition under the ordinary-gap and total-variation interpretations. The source's undefined conditional-gap, second-order, and comparison-language clauses are not assigned additional meanings. A closed Lean 4 certificate proves the stochastic, spectral, and mixing-time bridges used in the argument.

1 Original statement and precise scope

The original statement is reproduced in both supplied languages.

> **English.** Definition: Cutoff means that for a family of random walks the ratio of total mixing time to mixing window tends to infinity. Conjecture: Lazy random walks on abelian groups exhibit cutoff if and only if the second-order behavior of the conditional spectral gap over irreducible characters of the generating support satisfies an explicit criterion: the ratio of the multiplicity of the minimal nontrivial eigenvalue to the spectral gap tends to infinity; the criterion is completely characterized by Diaconis–Saloff-Coste type second-eigenvalue comparison.
>
> **中文**。定义：cutoff 指随机游走族的总混合时间与混合窗之比趋于无穷。猜想：交换群上惰性随机游走出现 cutoff 当且仅当其生成支撑的不可约特征条件谱隙的二阶行为满足显式判据: 最小非平凡特征值的重数与谱隙之比趋于无穷; 判据由 Diaconis–Saloff-Coste 型第二特征值比较完全刻画。

The explicit ratio criterion has a definite sufficient direction once the gap, eigenvalue convention, and mixing metric are specified. We refute that direction with the following scope.

1. **Gap and metric.** The gap is the ordinary spectral gap of the reversible transition operator. Mixing is measured in total variation from the uniform stationary law, with the maximum over starting states. The source does not itself name a metric or give a precise window definition. We state our definitions below and prove failure of both a genuine fixed-threshold width test and the quantified standard TV cutoff criterion.

2. **Two eigenvalue readings.** We compute separately the multiplicity of the smallest nontrivial eigenvalue of P and that of the smallest positive eigenvalue of $I-P$. The first is not silently replaced by the second. Both resulting ratios diverge in the same counterexample.

3. **Characters and support.** The relevant characters are those of the group generated by the support; the support itself need not be a group. Our support generates the entire state group. Its genuine complex group characters are the one-dimensional irreducible characters of this finite abelian group, and they form the complete spectral basis used here.

4. **Undefined prose.** The statement does not define a conditioning operation or a quantity called the "conditional spectral gap," a "second-order behavior," or a particular Diaconis–Saloff-Coste comparison assertion. We do not invent such definitions or claim a theorem about an unspecified comparison principle. If an additional nonstandard quantity or further restriction was intended, a clarified conjecture would be required. The result here concerns the explicit numerical sufficient criterion under the ordinary-gap and TV interpretations just stated.

The counterexample satisfies the narrower, familiar convention of exactly 1/2 holding and otherwise uniform choice from a symmetric nonidentity generating set. It uses neither variable holding probabilities nor a weighted generating set. Its state-space orders and ordinary mixing times both tend to infinity.

2 Definitions and the counterexample

Let $n \in \mathbb{N}$ and set $N = n + 4$. Thus $N \geq 4$ throughout, and every limit below is taken along this sequence as $n \to \infty$, equivalently $N \to \infty$. Write

$$G_N = (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/N\mathbb{Z}), \qquad S_N = \{(0,h) : h \neq 0\} \cup \{(1,0)\}.$$

Define a discrete-time random walk by adding independent increments with law

$$\mu_N(g) = \frac{1}{2}\mathbf{1}_{\{g=0\}} + \frac{1}{2N}\mathbf{1}_{\{g\in S_N\}}. \tag{1}$$

Its transition matrix and the candidate stationary law are

$$P_N(x,y) = \mu_N(y-x), \qquad \pi_N(y) = \frac{1}{2N}. \tag{2}$$

Powers P_N^t always mean the usual powers of this matrix, with $P_N^0 = I$.

Definition 2.1. For $t \in \mathbb{N}$ define

$$d_N(t) = \max_{x\in G_N} \frac{1}{2} \sum_{y\in G_N} \left|P_N^t(x,y) - \pi_N(y)\right|.$$

For $\varepsilon > 0$, the mixing time is the genuine least threshold time

$$t_N(\varepsilon) = \min\{t \in \mathbb{N} : d_N(t) \leq \varepsilon\},$$

once nonemptiness of this set has been proved. The ordinary mixing time and the quarter-to-three-quarter width are

$$T_N = t_N(1/4), \qquad W_N = t_N(1/4) - t_N(3/4).$$

The subtraction and ratios involving these times are taken in $\mathbb{R}$. The standard quantified total-variation cutoff criterion used here is

$$\text{for every } 0 < \varepsilon < \frac{1}{2}, \qquad \frac{t_N(\varepsilon)}{t_N(1-\varepsilon)} \longrightarrow 1. \tag{3}$$

For each fixed $0 < \varepsilon < 1/2$, the denominator is positive once $N > 1/(2\varepsilon)$, since $d_N(0) = 1 - 1/(2N) > 1 - \varepsilon$. Thus these asymptotic ratios are eventually defined; a convention at finitely many initial indices does not affect their limits. Lean uses total real division.

The fixed-tolerance ratio convention in (3) is standard; see Levin's introductory definition, slide 23 of [1]. That reference is cited only to identify the convention. The construction and proof below were derived independently before consulting it, and no proof content is taken from it. We use non-strict threshold inequalities in the definition of t_N, consistently with the Lean development.

Theorem 2.2. *The matrices in (2) give a growing family of irreducible, reversible, exactly-half-lazy uniform Cayley walks. Put*

$$q = 1 - \frac{1}{N}, \qquad r = \frac{1}{2}, \qquad s = \frac{1}{2} - \frac{1}{N}. \tag{4}$$

Their complete transition spectrum, with algebraic multiplicity, is

$$1\ (1), \qquad q\ (1), \qquad r\ (N-1), \qquad s\ (N-1).$$

The ordinary spectral gap is $\delta_N = 1/N$. The literal minimum nontrivial-transition multiplicity and the minimum positive $I - P_N$ multiplicity satisfy, respectively,

$$\frac{N-1}{\delta_N} = N(N-1) \longrightarrow \infty, \qquad \frac{1}{\delta_N} = N \longrightarrow \infty. \tag{5}$$

Nevertheless

$$d_N(0) = 1 - \frac{1}{2N}, \qquad d_N(t) = \frac{1}{2}q^t \quad (t \geq 1). \tag{6}$$

All positive-tolerance mixing times are finite and attained, and

$$t_N(3/4) = 1, \qquad T_N \geq \frac{N}{2}, \qquad W_N = T_N - 1 > 0, \qquad \frac{T_N}{W_N} \longrightarrow 1. \tag{7}$$

In particular $T_N \to \infty$, the ratio T_N/W_N does not tend to infinity, and the family does not satisfy (3).

The remaining sections prove every assertion of this theorem directly.

3 The support and the stochastic walk

The horizontal part of S_N has $N - 1$ elements and is disjoint from the single vertical element $(1, 0)$. Hence $|S_N| = N$ and $0 \notin S_N$. Negation preserves the horizontal part, while $-(1, 0) = (1, 0)$; thus $S_N = -S_N$. Every $(a, b) \in G_N$ is

$$(a, b) = (a, 0) + (0, b).$$

Each summand is either zero or belongs to S_N, because a is either zero or one. Consequently the subgroup generated by S_N is all of G_N. The group order is $|G_N| = 2N = 2(n + 4) \to \infty$.

More explicitly, from (a, b) the chain has probabilities

destination	probability
(a, b)	$1/2$
$(a, d),\ d \neq b$	$1/(2N)$ for each d
$(a + 1, b)$	$1/(2N)$.

All other transition probabilities are zero. The probabilities are nonnegative and each row sums to $1/2 + N/(2N) = 1$. Symmetry of the support and increment law gives $P_N(x, y) = P_N(y, x)$, so the column sums are also one. The law π_N is nonnegative, has total mass one, and satisfies

$$\sum_x \pi_N(x) P_N(x, y) = \frac{1}{2N} \sum_x P_N(x, y) = \pi_N(y).$$

This symmetry also proves detailed balance with respect to π_N.

For any $x = (a, b)$ and $y = (c, d)$, choose $z = (a, d)$. Then $P_N(x, z) > 0$, since x and z have the same first coordinate, and $P_N(z, y) > 0$, since z and y have the same second coordinate and their first coordinates are either equal or differ by one. Therefore

$$P_N^2(x, y) = \sum_z P_N(x, z) P_N(z, y) > 0.$$

The chain is irreducible. Its diagonal entries are exactly $1/2$, so it is lazy and aperiodic. Finally, products of nonnegative stochastic matrices are stochastic: summing a row of AB gives $\sum_z A(x, z) \sum_y B(z, y) = 1$. Thus every actual matrix power P_N^t is a probability kernel, as required in Definition 2.1.

4 Complete character spectra and genuine multiplicities

4.1 Characters form the entire eigenbasis

A complex character of a finite additive group is a map χ satisfying $\chi(0) = 1$ and $\chi(u + v) = \chi(u)\chi(v)$. Its values have modulus one: if $ku = 0$, then $\chi(u)^k = 1$. The N characters of $\mathbb{Z}/N\mathbb{Z}$ are $\chi_j(b) = \exp(2\pi i j b/N)$, $0 \le j < N$; indeed a character is determined by its value at one, which must be an Nth root of unity. Similarly the two characters of $\mathbb{Z}/2\mathbb{Z}$ are $a \mapsto (-1)^{\epsilon a}$ for $\epsilon \in \{0, 1\}$.

Every character of G_N factors uniquely as

$$\psi_{\epsilon,\chi}(a, b) = (-1)^{\epsilon a} \chi(b). \tag{8}$$

To see completeness of this list as characters, restrict any character ψ to the two factors and use $\psi(a, b) = \psi(a, 0)\psi(0, b)$. Uniqueness follows from the same restrictions.

For a nontrivial character η of any finite group H, choose h_0 with $\eta(h_0) \neq 1$. Translation of the sum gives

$$\sum_{h \in H} \eta(h) = \sum_{h \in H} \eta(h + h_0) = \eta(h_0) \sum_{h \in H} \eta(h),$$

so the sum is zero. For the trivial character it is $|H|$. For distinct characters, their quotient is nontrivial; this identity gives orthogonality under the usual complex inner product on functions on H. The $2N$ distinct characters in (8) are therefore orthogonal nonzero vectors in the $2N$-dimensional space $\mathbb{C}^{G_N}$. They form a basis. In particular, no spectral modes are omitted.

4.2 Action of the actual transition matrix

For a character ψ and a state x, the matrix from (2) satisfies

$$\begin{aligned}(P_N\psi)(x) &= \sum_y \mu_N(y - x)\psi(y) \\ &= \sum_g \mu_N(g)\psi(x + g) = \psi(x) \sum_g \mu_N(g)\psi(g).\end{aligned}$$

Thus the actual eigenvalue of $\psi_{\epsilon,\chi}$ is

$$\lambda_{\epsilon,\chi} = \frac{1}{2} + \frac{1}{2N}\left(\sum_{h\neq 0}\chi(h) + (-1)^{\epsilon}\right). \tag{9}$$

The preceding sum identity and $\chi(0) = 1$ give

$$\sum_{h\neq 0}\chi(h) = \begin{cases} N-1, & \chi \text{ trivial}, \\ -1, & \chi \text{ nontrivial}. \end{cases}$$

Substitution into (9) yields

First-factor character	Second-factor character	Eigenvalue	Multiplicity
Trivial	Trivial	1	1
Nontrivial	Trivial	$q = 1 - 1/N$	1
Trivial	Nontrivial	$r = 1/2$	$N-1$
Nontrivial	Nontrivial	$s = 1/2 - 1/N$	$N-1$

Because this is a complete basis of eigenvectors, these counts are the algebraic multiplicities of the actual operator. Equivalently its characteristic polynomial over $\mathbb{C}$ is

$$\det(XI - P_N) = (X-1)(X-q)(X-r)^{N-1}(X-s)^{N-1}. \tag{10}$$

For $N \geq 4$ the values are strictly ordered:

$$0 < s < r < q < 1. \tag{11}$$

Consequently the only stationary character is the trivial one, and the ordinary spectral gap is

$$\delta_N = 1 - \max\{\lambda : \lambda \in \operatorname{spec}(P_N),\ \lambda \neq 1\} = 1 - q = \frac{1}{N}.$$

All eigenvalues are positive, so using the largest absolute nontrivial eigenvalue produces the same gap in this family.

4.3 The distinct $I - P_N$ interpretation

The same complete character basis diagonalizes the actual complementary operator $L_N = I - P_N$, with eigenvalue $1 - \lambda_{\epsilon,\chi}$ on each character. Its characteristic polynomial is

$$\det(XI - L_N) = X\big(X-(1-q)\big)\big(X-(1-r)\big)^{N-1}\big(X-(1-s)\big)^{N-1}. \tag{12}$$

Thus its complete spectrum, with multiplicity, is

$$0\ (1), \qquad \frac{1}{N}\ (1), \qquad \frac{1}{2}\ (N-1), \qquad \frac{1}{2}+\frac{1}{N}\ (N-1).$$

The smallest positive eigenvalue is precisely $\delta_N = 1/N$, and its algebraic multiplicity is one. In contrast, the smallest nontrivial *transition* eigenvalue is s, with multiplicity $N-1$. Distinctness of the roots in (10) and (12) makes these exact root multiplicities, rather than merely lower bounds supplied by some eigenvectors. Both claims in (5) now follow. Their divergence follows from $N \to \infty$, since $N(N-1) \geq N$ for $N \geq 4$.

5 An identity for the actual matrix powers

We derive the transition probabilities directly by matrix multiplication. This also supplies a proof of the bridge between the stochastic walk and the scalar expressions used for its mixing analysis.

For $x = (a, b)$ and $y = (c, d)$ put

$$U = \mathbf{1}_{\{a=c\}}, \qquad V = \mathbf{1}_{\{b=d\}}, \qquad \sigma = 2U - 1 \in \{1, -1\}.$$

For four real numbers A, B, C, D, let

$$F(A, B, C, D)(x, y) = A + BU + CV + D\mathbf{1}_{\{x=y\}}.$$

This is an auxiliary matrix, not a new definition of the walk. Write $c_N = 1/(2N)$. The original transition kernel has the decomposition

$$P_N(x, y) = s\mathbf{1}_{\{x=y\}} + c_N U + c_N V. \tag{13}$$

When $x = y$ the right side is $s + 2c_N = 1/2$; in the two possible nonidentity neighbor cases it is c_N; otherwise it is zero. This verifies (13) directly from (2).

The coordinate sums of F are

$$\sum_{d \in \mathbb{Z}/N\mathbb{Z}} F(A, B, C, D)(x, (c, d)) = NA + C + (NB + D)\mathbf{1}_{\{a=c\}}, \tag{14}$$

$$\sum_{c \in \mathbb{Z}/2\mathbb{Z}} F(A, B, C, D)(x, (c, d)) = 2A + B + (2C + D)\mathbf{1}_{\{b=d\}}. \tag{15}$$

For an arbitrary matrix M, (13) gives

$$(MP_N)(x, y) = sM(x, y) + c_N \sum_{v \in \mathbb{Z}/N\mathbb{Z}} M(x, (c, v)) + c_N \sum_{u \in \mathbb{Z}/2\mathbb{Z}} M(x, (u, d)).$$

Using (14)–(15), and $s + c_N N + 2c_N = 1$, $s + c_N N = q$, $s + 2c_N = r$, we obtain

$$F(A, B, C, D)P_N = F\big(A + c_N(B + C),\ qB + c_N D,\ rC + c_N D,\ sD\big). \tag{16}$$

Define coefficients, for all integers $t \geq 0$, by

$$A_t = \frac{1 - q^t - r^t + s^t}{2N}, \qquad B_t = \frac{q^t - s^t}{N},$$
$$C_t = \frac{r^t - s^t}{2}, \qquad D_t = s^t.$$

At time zero these are $(0, 0, 0, 1)$, so $F(A_0, B_0, C_0, D_0) = I$. They satisfy precisely the four recurrences

$$A_{t+1} = A_t + c_N(B_t + C_t), \qquad B_{t+1} = qB_t + c_N D_t,$$
$$C_{t+1} = rC_t + c_N D_t, \qquad D_{t+1} = sD_t.$$

For example, $q - 1/2 = s$ gives the recurrence for B_t, and $r - 1/N = s$ gives that for C_t. For the constant coefficient,

$$A_{t+1} - A_t = \frac{(1 - q)q^t + (1 - r)r^t - (1 - s)s^t}{2N} = c_N \left(\frac{q^t - s^t}{N} + \frac{r^t - s^t}{2} \right),$$

using $1 - q = 1/N$, $1 - r = 1/2$, and $1 - s = 1/2 + 1/N$. The recurrence for D_t is immediate. Induction with (16) therefore proves $P_N^t = F(A_t, B_t, C_t, D_t)$ for every $t \geq 0$.

Rearranging this identity using $\sigma = 2U - 1$ proves the promised formula:

Proposition 5.1. *For every $t \in \mathbb{N}$ and every pair of states $x = (a, b)$, $y = (c, d)$,*

$$P_N^t(x, y) = \frac{1 + \sigma q^t + (NV - 1)(r^t + \sigma s^t)}{2N}. \tag{17}$$

In particular this formula holds at $t = 0$ and equals the identity kernel there.

One useful marginal follows either from the coefficient formula or by summing (17) over d:

$$\sum_{d \in \mathbb{Z}/N\mathbb{Z}} P_N^t(x, (c, d)) = \frac{1 + \sigma q^t}{2}, \tag{18}$$

because $\sum_d (N\mathbf{1}_{\{b=d\}} - 1) = 0$.

6 Exact total variation, including time zero

The signs inside the absolute values are essential. We establish them before summing; no norm approximation or spectral bound replaces the actual TV distance.

Lemma 6.1. *For each integer $t \geq 1$,*

$$r^t + s^t \leq q^t, \qquad (N - 1)(r^t - s^t) \leq q^t.$$

Proof. We have $q = r + s$ and $r, s \geq 0$. The first inequality follows by expanding $(r + s)^t$ and retaining its two endpoint terms. For the second, use $(N - 1)(r - s) = q$ and the difference-of-powers identity:

$$(N - 1)(r^t - s^t) = q \sum_{j=0}^{t-1} r^{t-1-j} s^j \leq q(r + s)^{t-1} = q^t.$$

The middle inequality holds because every binomial coefficient in $(r+s)^{t-1}$ is at least one and all monomials are nonnegative. This also covers $t = 1$, for which the sum has its single term equal to one. The Lean proof establishes the same inequalities by elementary induction. □

Fix a starting state $x = (a, b)$ and $t \geq 1$. Subtracting $1/(2N)$ from (17) gives the numerator

$$\sigma q^t + (NV - 1)(r^t + \sigma s^t).$$

In the block $c = a$ this is nonnegative: for $V = 1$ both summands are nonnegative, while for $V = 0$ it is $q^t - r^t - s^t \geq 0$. In the other block it is nonpositive: for $V = 1$ it is $-q^t + (N - 1)(r^t - s^t) \leq 0$, while for $V = 0$ it is $-q^t - r^t + s^t \leq 0$, since $s \leq r$. These signs remain valid after division by the positive number $2N$.

By (18), the sum of the stationary deviations in the starting block is

$$\frac{1 + q^t}{2} - \frac{1}{2} = \frac{q^t}{2}.$$

In the other block it is $-q^t/2$. The absolute deviations therefore sum to q^t, and

$$\frac{1}{2} \sum_y \left| P_N^t(x, y) - \pi_N(y) \right| = \frac{q^t}{2} \qquad (t \geq 1). \tag{19}$$

This equality holds for every starting state x, so it also gives their maximum. At time zero the row of P_N^0 is a point mass, and direct summation instead gives

$$\frac{1}{2} \left(1 - \frac{1}{2N} + (2N - 1) \frac{1}{2N} \right) = 1 - \frac{1}{2N}.$$

Together these prove (6).

7 True mixing times and the absence of cutoff

7.1 Existence and least-time properties

For fixed N, $0 < q < 1$, so $q^t \to 0$. For completeness, this follows without logarithmic asymptotics: set $a = q^{-1} - 1 > 0$. Bernoulli's inequality gives $(1+a)^t \geq 1 + ta$, and hence

$$0 < q^t \leq \frac{1}{1+ta} \longrightarrow 0.$$

By (19), for every $\varepsilon > 0$ some positive integer t satisfies $d_N(t) \leq \varepsilon$. The threshold set is nonempty and, by well-ordering of $\mathbb{N}$, has a minimum. In particular,

$$d_N(t_N(\varepsilon)) \leq \varepsilon, \qquad u < t_N(\varepsilon) \implies d_N(u) > \varepsilon. \tag{20}$$

These are properties of the walk's actual least mixing times, not an assumed or substituted scalar sequence.

7.2 Two exact thresholds and a divergent mixing time

Since $N \geq 4$,

$$d_N(0) = 1 - \frac{1}{2N} > \frac{3}{4}, \qquad d_N(1) = \frac{q}{2} < \frac{1}{2} < \frac{3}{4}.$$

Thus $t_N(3/4) = 1$ exactly. Likewise $T_N = t_N(1/4)$ is positive, so attainment in (20) and (19) imply $q^{T_N} \leq 1/2$. Bernoulli's inequality, now with increment $-1/N$, gives

$$1 - \frac{T_N}{N} \leq \left(1 - \frac{1}{N}\right)^{T_N} = q^{T_N} \leq \frac{1}{2}.$$

It follows that

$$T_N \geq \frac{N}{2} \geq 2. \tag{21}$$

In particular $T_N \to \infty$, while the high-tolerance mixing time stays equal to one.

7.3 The actual transition width and quantified cutoff

Our width is the difference between the genuine threshold times:

$$W_N = T_N - t_N(3/4) = T_N - 1 \geq 1.$$

Consequently

$$\frac{T_N}{W_N} = \frac{T_N}{T_N - 1} = 1 + \frac{1}{T_N - 1} \longrightarrow 1. \tag{22}$$

For every $N \geq 4$ this ratio is greater than one and at most two. In particular, it cannot tend to infinity. The time needed to move from TV distance $3/4$ to $1/4$ has the same order as the ordinary mixing time; it is not a window of vanishing relative width.

The quantified standard cutoff test fails independently at the single allowed tolerance $\varepsilon = 1/4$:

$$\frac{t_N(1/4)}{t_N(1 - 1/4)} = \frac{T_N}{1} = T_N \longrightarrow \infty,$$

whereas (3) would require this ratio to tend to one. This proves all remaining assertions of Theorem 2.2.

Corollary 7.1. *Under the ordinary spectral-gap and standard total-variation meanings, divergence of the proposed multiplicity-to-gap ratio is not sufficient for cutoff on lazy abelian-group random walks. This holds whether "minimal nontrivial eigenvalue" means the smallest nontrivial eigenvalue of P_N or the smallest positive eigenvalue of $I - P_N$.*

Proof. A universal sufficient condition would apply to every family in the asserted class, hence to (2). The support, group, and stochastic properties above prove that this is such a family, even with exact half laziness and uniform supported steps. Equation (5) proves both candidate antecedents. Equations (22) and (3)'s failure prove the corresponding conclusions false. Thus this one fully specified witness refutes the universal sufficient direction. Refuting that direction is enough to refute the proposed iff under these meanings; the necessary direction need not be decided. □

8 Correspondence with the frozen Lean certificate

The formalization uses the same index $N = n+4$, state group, support, transition kernel, and threshold conventions as this report. Its principal definitions are summarized below. Every spectral and mixing formula used as a conclusion in the preceding proof is proved from these definitions.

Lean definition	Mathematical object
`State`, `support`	G_N and the actual generating set S_N.
`jumpLaw`, `transition`	μ_N and $P_N(x, y) = \mu_N(y - x)$.
`tvFrom`, `worstTV`	The absolute-value finite TV sum from each start and its finite maximum.
`Characters`	Actual complex additive characters of G_N.
`fourierEigenvalue`	The Fourier sum of the actual increment law, proved to be its matrix eigenvalue.
`complexTransition`	Entrywise complexification of the already defined real matrix P_N.
`spectralGap`	One minus the finite maximum nontrivial character eigenvalue. All values are proved real.
`minTransitionEigenvalue`	The finite minimum of the actual nontrivial transition eigenvalues.
`minTransitionMultiplicity`	Root multiplicity in the actual transition characteristic polynomial at that minimum.
`complexComplement`	The actual matrix $I - P_N$.
`gapModeMultiplicity`	Root multiplicity in the actual complement characteristic polynomial at the proved ordinary gap.
`mixingTime`	The natural-number infimum of the actual TV threshold set, with nonemptiness, attainment, and minimality proved.
`totalTime`, `window`	Real-valued T_N and $T_N - t_N(3/4)$.
`HasTVCutoff`	The quantified condition (3).

The proof modules divide the obligations as follows.

- `Basic.lean` gives the actual objects, normalizations, and elementary inequalities. `Walk.lean` proves support generation, the normalized uniform Cayley formula, stochasticity, stationary mass, stochastic powers, exact holding, and positive two-step probabilities.
- `Spectrum.lean` constructs the equivalence between factor characters and all group characters. It proves actual matrix action and diagonalization in `AddChar.complexBasis`, both characteristic polynomials, complete root classifications, finite extrema, and the genuine algebraic multiplicities.

- `Powers.lean` proves the matrix recurrence and actual power formula, the sign inequalities, and the finite absolute-value TV sums, including time zero and the maximum over all starting states.
- `MixingBase.lean` defines the actual least-threshold times and proves their properties and limits. Its intermediate lemmas explicitly take the exact TV profile as a parameter. In the final theorem every such parameter is discharged with `worstTV_eq`, the identity proved in `Powers.lean`; it is not an assumption of the final result.
- `Counterexample.lean` assembles the closed certificate and four sufficient-condition refutations. `Audit.lean` requests the axiom lists of the five capstone theorems. The top-level library imports the counterexample module.

All capstones lie in the namespace `Conjecture1421`:

1. `verified_counterexample` proves `CounterexampleCertificate`, a proposition exposing the actual walk, group-order divergence, support, probability, character, spectral, matrix-power, TV, minimum-time, window, and cutoff obligations together. It has no mathematical premise.
2. `literal_sufficient_criterion_false` negates the implication from the minimum-transition multiplicity ratio's divergence to `HasTVCutoff` for this actual family.
3. `gap_mode_sufficient_criterion_false` does the same for the minimum positive eigenvalue of the actual $I - P_N$.
4. `literal_window_sufficient_criterion_false` negates the implication from the first ratio's divergence to divergence of the actual time/window ratio.
5. `gap_mode_window_sufficient_criterion_false` does the same for the second ratio.

The four corollaries are deliberately statements about this fully specified witness; they do not pretend to encode every undefined phrase of the original or an entire class of arbitrary random walks. The closed certificate establishes membership in the proposed class and the proved antecedents. Corollary 7.1 explains the logical step from this witness to failure of universal sufficiency. No profile assumption, spectral formula, multiplicity, or mixing-time bound remains as a hypothesis of any capstone.

9 Reproduction and supplemental verification

9.1 Pinned formal project

The project requests Lean `leanprover/lean4:v4.19.0` and Mathlib `v4.19.0`. Its dependency manifest pins Mathlib to

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The manifest also pins all transitive dependencies. Warnings are treated as errors in the project configuration. From the directory containing `lakefile.toml`, with the pinned toolchain and dependencies available, the reproduction commands are

```
lake build
lake env lean -DwarningAsError=true Conjecture1421/Audit.lean
```

The audit command imports the actual capstones rather than proving an unrelated sample theorem. The author-stage fresh direct Lean replay of all eight modules succeeded with

warnings treated as errors. Its capstone axiom output consists only of the standard Lean logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The source has no admitted proof, added axiom, native-decision proof, or custom unsafe proof dependency.

The separate isolated engineering run completed a source-based Lake build and eight strict module replays successfully. Its audit counted 342 owned declarations, 292 safe owned declarations, 31,861 declarations in the combined dependency closure, and 3,701 imported modules. The five capstones again had only the three standard axioms listed above. An independent finite-regression rerun reproduced the recorded output exactly. A positive control, five negative controls, and seven build-harness hardening fixtures also completed as expected. The independent engineering review passed after explicitly classifying all 50 generated unsafe runtime helpers (42 definitions and eight compiler placeholder axioms). None occurs in any safe owned proof closure or in any of the five capstones. The raw driver status, which leaves this classification pending, is preserved alongside the completed separate classification and verdict. All 62 command receipts are retained, including six intentional nonzero negative-control results; all stderr streams are empty.

These checks are local verification, not maintainer acceptance. Cached generic library artifacts and pinned source bytes were identified separately; this is not a full Mathlib rebuild or proof of source-to-cache provenance. Lean was not bootstrapped. The dependency audit and graph consistency checks are not a second proof kernel; the fresh source compilations supply kernel checking relative to the supplied runtime and imports. PATH network wrappers are not operating-system isolation. Complete source and configuration review complements these checks. Detailed identities, controls, and limits are preserved in the accompanying verification records. The finite evidence below remains supplementary to the mathematical proof.

9.2 Exact finite regression

The supplemental `exact_regression.py` uses exact rational arithmetic. It independently constructs increments and transition probabilities and evolves distributions by direct multiplication. Its recorded run covered

$$N \in \{4, 5, 6, 7, 8, 11, 16\}, \qquad t = 0, 1, \ldots, 24,$$

all starting states, and 56,700 actual kernel entries. It checked the power identities, actual absolute-value TV distances, least threshold times, and an explicit real eigenbasis with its rank and eigenvalue action. The run also included three intentional negative controls. To reproduce these supplemental checks, run `python3 exact_regression.py` from the directory containing the script.

These finite checks are corroborating arithmetic evidence only. They do not establish a universal-in-N or universal-in-t identity, an asymptotic limit, or a Lean theorem. The proofs in this report and the closed formal certificate supply those claims without using the regression output as an assumption.

10 Conclusion and scope of the refutation

The actual family (2) has growing finite abelian state groups, uniform symmetric supported steps, exact half laziness, and diverging ordinary mixing times. Its full spectra show that both proposed multiplicity-to-ordinary-gap ratios tend to infinity. Its actual total-variation profile and least threshold times show simultaneously that standard TV cutoff fails and the genuine time/window ratio tends to one. This refutes the explicit sufficient direction, and therefore the explicit iff, under the stated ordinary-gap and total-variation meanings.

The interpretation limits remain part of the result. The source does not specify a mixing metric or an exact window convention, and it does not define a conditional spectral gap, second-order behavior, or a particular Diaconis–Saloff-Coste comparison characterization. We have not formalized those undefined phrases or claimed a result about every possible mixing metric or an unstated additional hypothesis. Both natural meanings of "minimal nontrivial eigenvalue" have, however, been handled by actual operator spectra and actual algebraic multiplicities, with no conflation between the two.

References

[1] David A. Levin, *Cutoff for Markov Chains*, LMS–EPSRC Durham Symposium, 2017, slide 23 (PDF page 23). University of Oregon slide deck. Cited only for the standard cutoff convention, after the independent derivation of this counterexample.